

Manufacturing Equipment Output Prediction

Predicting hourly output (Parts_Per_Hour) of a plastic injection molding machine from process and operational data

Project Report

Source notebook: Manufacturing_Output_Prediction_Polished.ipynb

Dataset: manufacturing_dataset_1000_samples.csv (1,000 rows, 19 columns)

Model: scikit-learn Pipeline (imputation, scaling, one-hot encoding) + LinearRegression

Evaluation: 80/20 train-test split, random_state = 42

0.906

Test R²

2.716

Test MAE (parts/hr)

3.509

Test RMSE (parts/hr)

0.018

Train-test R² gap

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1. Executive Summary

This report documents a complete regression workflow that predicts the hourly production output (**Parts_Per_Hour**) of a plastic injection molding machine. The dataset has 1,000 hourly records with 18 input variables covering machine settings, environmental conditions, operator experience, maintenance and categorical production factors. A scikit-learn pipeline imputes, scales and encodes the inputs before fitting an ordinary least squares Linear Regression model.

- **The model explains about 91% of output variation on unseen data.** Test R^2 is 0.906 against 0.924 on training data, so overfitting is minimal.
- **Typical prediction error is under 3 parts per hour.** Test MAE is 2.716 against an average output of 29.3 parts per hour.
- **Cycle duration dominates output.** Cycle_Time ($r = -0.819$) and Total_Cycle_Time ($r = -0.793$) are the strongest correlates, and Total_Cycle_Time has the largest model coefficient (-10.06).
- **Operator experience and efficiency help.** Both have moderate positive correlations with output ($r = 0.301$ and 0.287).
- **Two data features limit the linear fit.** Output is floored at about 15 parts per hour for roughly 200 records, and this produces a visible residual pattern at low predicted values.

2. Introduction and Dataset Overview

2.1 Project Objective

Injection molding throughput depends on many interacting settings. Planners need a reliable estimate of hourly output to schedule production and to spot machines that under-perform. The objective of this project is to predict **Parts_Per_Hour** from operating parameters, environmental conditions, operator experience, maintenance information and categorical manufacturing factors, and to turn the results into practical recommendations.

2.2 Dataset at a Glance

Property	Value
Records	1,000 hourly observations starting 2023-01-01 00:00
Columns	19 (1 timestamp, 13 numerical inputs, 4 categorical inputs, 1 target)
Target	Parts_Per_Hour (continuous, 5.0 to 68.6)
Missing values	60 cells in 3 columns (2% of each affected column)
Duplicate rows	0

2.3 Variable Dictionary

Variable	Type	Business Meaning
Injection_Temperature	float	Temperature at which material is injected into the mold
Injection_Pressure	float	Pressure used during material injection
Cycle_Time	float	Time required for a complete production cycle
Cooling_Time	float	Time spent cooling the molded part
Material_Viscosity	float	Measure of material flow characteristics
Ambient_Temperature	float	Temperature of the surrounding production environment
Machine_Age	float	Age of the manufacturing machine
Operator_Experience	float	Experience level of the operator
Maintenance_Hours	int	Maintenance time associated with the machine
Temperature_Pressure_Ratio	float	Ratio of injection temperature to injection pressure
Total_Cycle_Time	float	Combined cycle-related processing time
Efficiency_Score	float	Overall process or machine efficiency measure
Machine_Utilization	float	Degree to which the machine is being utilized
Shift	category	Production shift: Day, Evening or Night
Machine_Type	category	Machine category: Type_A, Type_B or Type_C
Material_Grade	category	Material grade: Economy, Standard or Premium
Day_of_Week	category	Day on which the record was captured
Timestamp	datetime	Date and time of the record (converted to Hour)
Parts_Per_Hour	float	Target variable: hourly production output

3. Summary Statistics and Data Quality

3.1 Numerical Summary Statistics

Variable	Count	Mean	Std	Min	25%	Median	75%	Max
Injection_Temperature	1000	215.32	12.00	180.0	207.2	215.3	222.8	300.0
Injection_Pressure	1000	116.08	14.67	80.0	105.9	115.95	125.93	150.0
Cycle_Time	1000	35.85	8.35	16.3	28.8	36.85	45.0	60.0
Cooling_Time	1000	11.92	2.30	8.0	10.28	11.9	13.5	19.9
Material_Viscosity	980	251.63	73.35	104.6	200.9	242.7	295.3	1000.0
Ambient_Temperature	980	22.94	2.77	18.0	20.8	22.9	25.1	28.0
Machine_Age	1000	7.86	3.90	1.0	4.7	7.9	11.1	15.0
Operator_Experience	980	30.80	27.68	1.0	9.8	22.1	43.43	120.0
Maintenance_Hours	1000	50.58	16.01	26	45	50	55	500
Temp_Pressure_Ratio	1000	1.886	0.274	1.286	1.684	1.850	2.044	2.843
Total_Cycle_Time	1000	47.74	8.67	24.6	41.0	48.55	55.3	64.9
Efficiency_Score	1000	0.193	0.174	0.006	0.061	0.139	0.274	0.840
Machine_Utilization	1000	0.357	0.196	0.001	0.193	0.356	0.520	0.755
Parts_Per_Hour	1000	29.30	11.96	5.0	17.5	28.2	38.0	68.6

3.2 Missing Values and Duplicates

Column	Missing	Share
Material_Viscosity	20	2.0%
Ambient_Temperature	20	2.0%
Operator_Experience	20	2.0%
All other columns	0	0.0%
Duplicate rows	0	-

Only three numerical columns contain gaps, each with exactly 20 missing values. The notebook fills them with the training-set median inside the pipeline, so the test data never influences the imputed values. No duplicate records were found, so no rows were removed.

3.3 Unusual Values in the Summary

Several maxima sit far above the upper quartile. These look like data-entry anomalies or rare extreme settings, and the notebook keeps them unchanged:

- **Maintenance_Hours** reaches 500 while 75% of records are at or below 55.
- **Material_Viscosity** reaches 1000 while the upper quartile is 295.3.
- **Injection_Temperature** reaches 300 while the upper quartile is 222.8.
- **Cycle_Time** has one value of 60 above an apparent cap at 45.

4. Target Variable Analysis

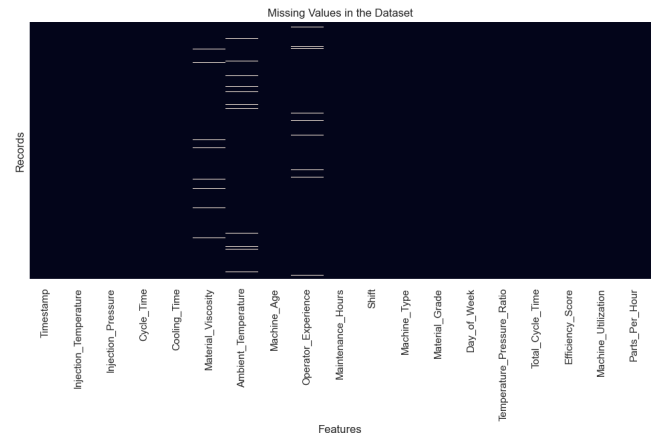


Figure 1. Missing-value map. Light marks show the scattered gaps in Material_Viscosity, Ambient_Temperature and Operator_Experience.

Statistic	Parts_Per_Hour
Count	1000
Mean	29.298
Std	11.955
Min	5.0
25%	17.5
Median	28.2
75%	38.0
Max	68.6

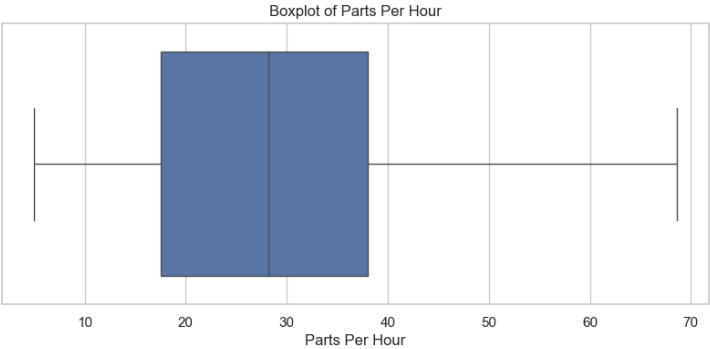


Figure 2. Boxplot of Parts_Per_Hour.

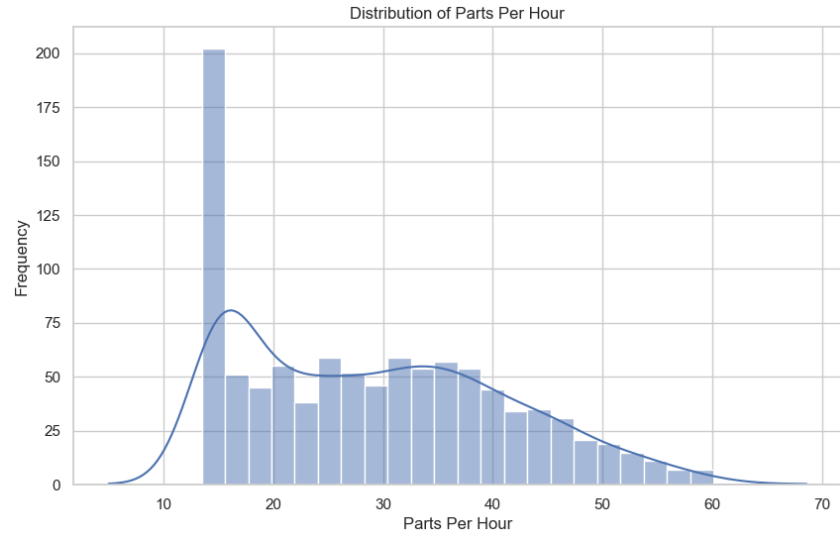


Figure 3. Distribution of Parts_Per_Hour with kernel density estimate.

Output ranges from 5 to 68.6 parts per hour with a mean of 29.3 and a median of 28.2. The distribution is right-skewed. A tall spike of about 200 records sits at roughly 15 parts per hour, which points to a recorded minimum rather than natural variation.

5. Numerical Feature Distributions

The histograms below cover all numerical variables. Most process settings are roughly symmetric around their means. Operator_Experience, Efficiency_Score and Parts_Per_Hour are right-skewed.

Histograms of Numerical Variables

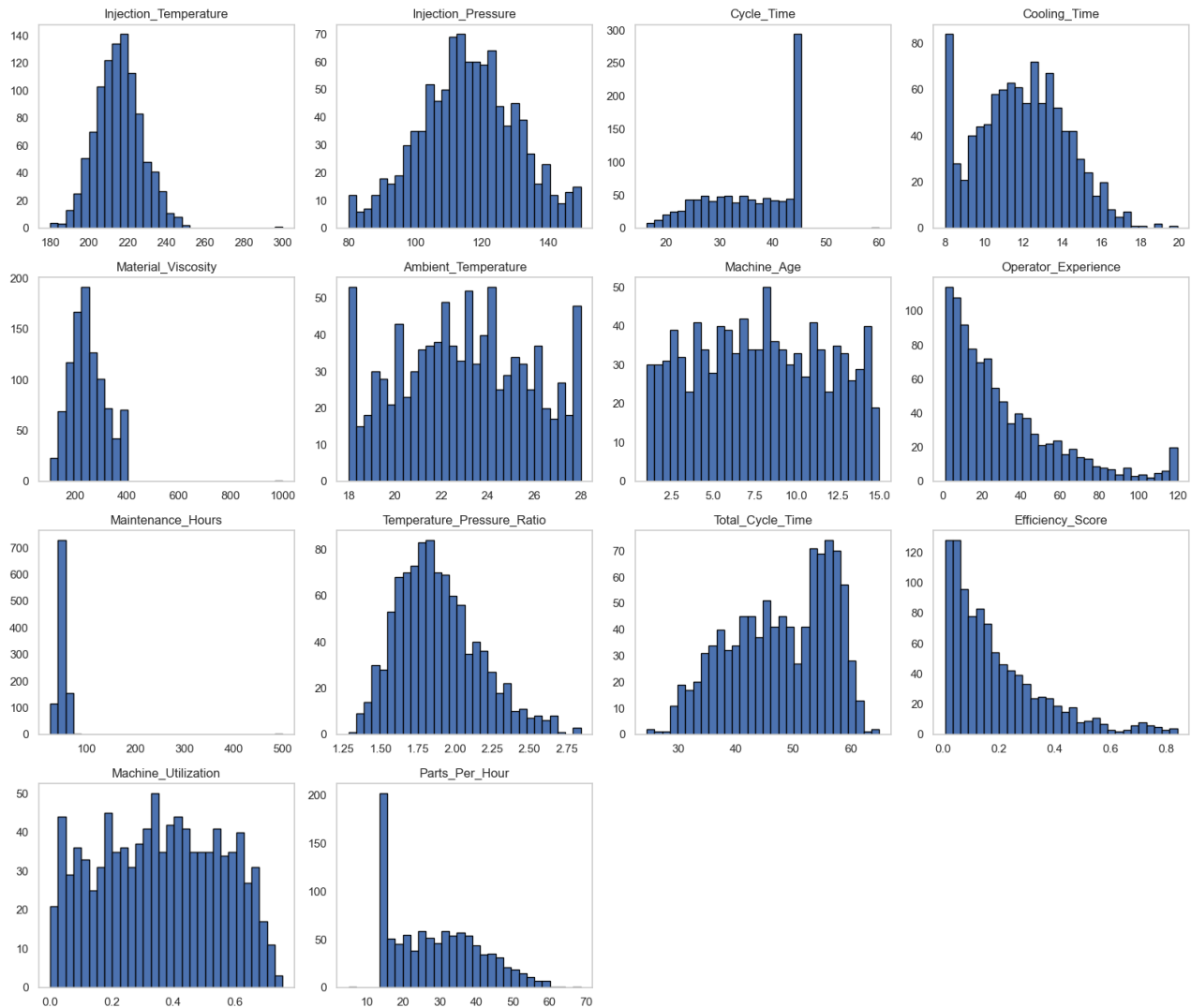


Figure 4. Histograms of all numerical variables (30 bins each).

- **Ceiling and floor effects are visible.** Cycle_Time piles up at 45 and Parts_Per_Hour piles up near 15.
- **Extreme values stretch some axes.** Maintenance_Hours and Material_Viscosity have single points far to the right of the main mass.

6. Correlation Analysis

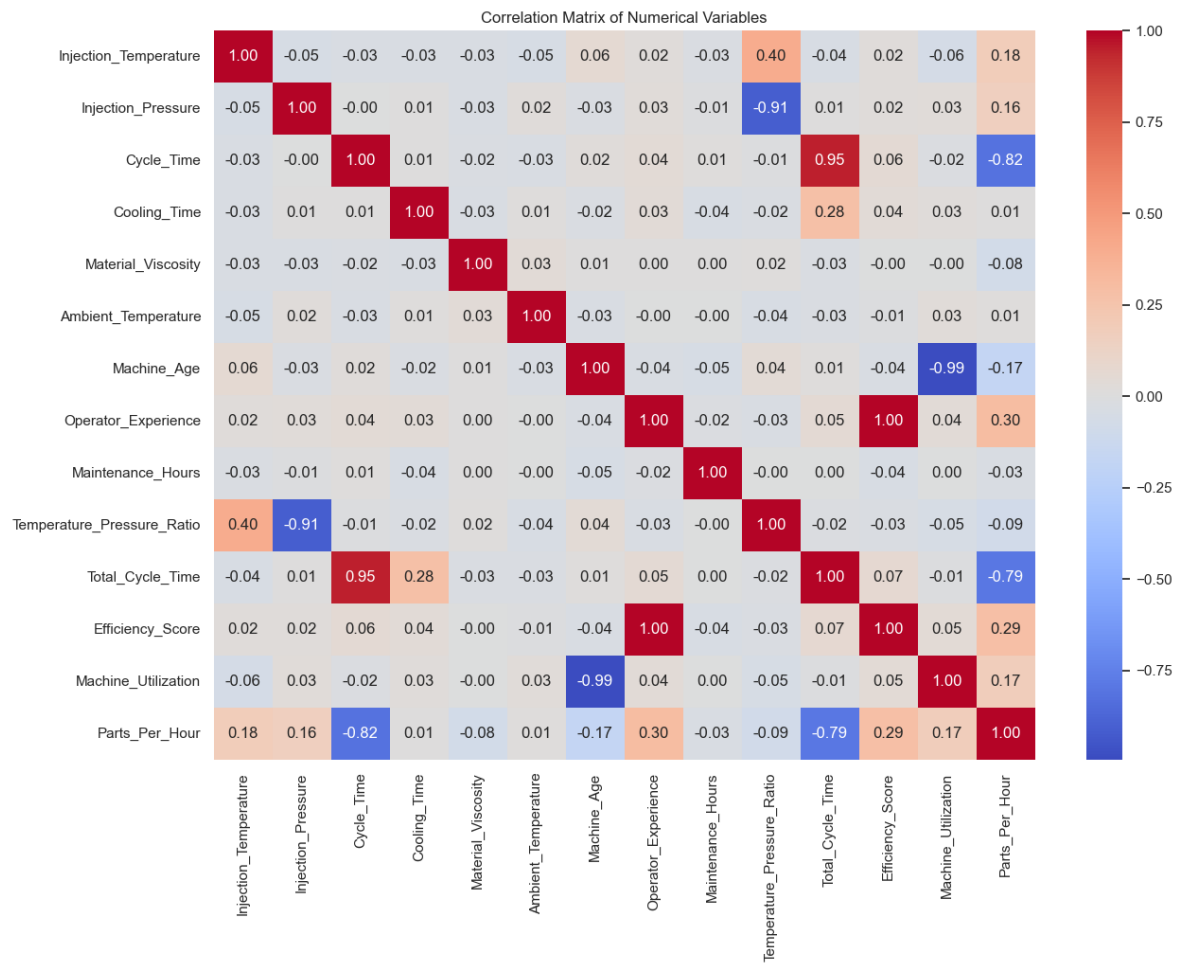


Figure 5. Pearson correlation matrix of numerical variables.

Feature	r with output	Feature	r with output
Operator_Experience	0.301	Maintenance_Hours	-0.033
Efficiency_Score	0.287	Material_Viscosity	-0.081
Injection_Temperature	0.184	Temp_Pressure_Ratio	-0.086
Machine_Utilization	0.175	Machine_Age	-0.175
Injection_Pressure	0.160	Total_Cycle_Time	-0.793
Ambient_Temperature	0.012	Cycle_Time	-0.819
Cooling_Time	0.006		

6. Correlation Analysis (continued)

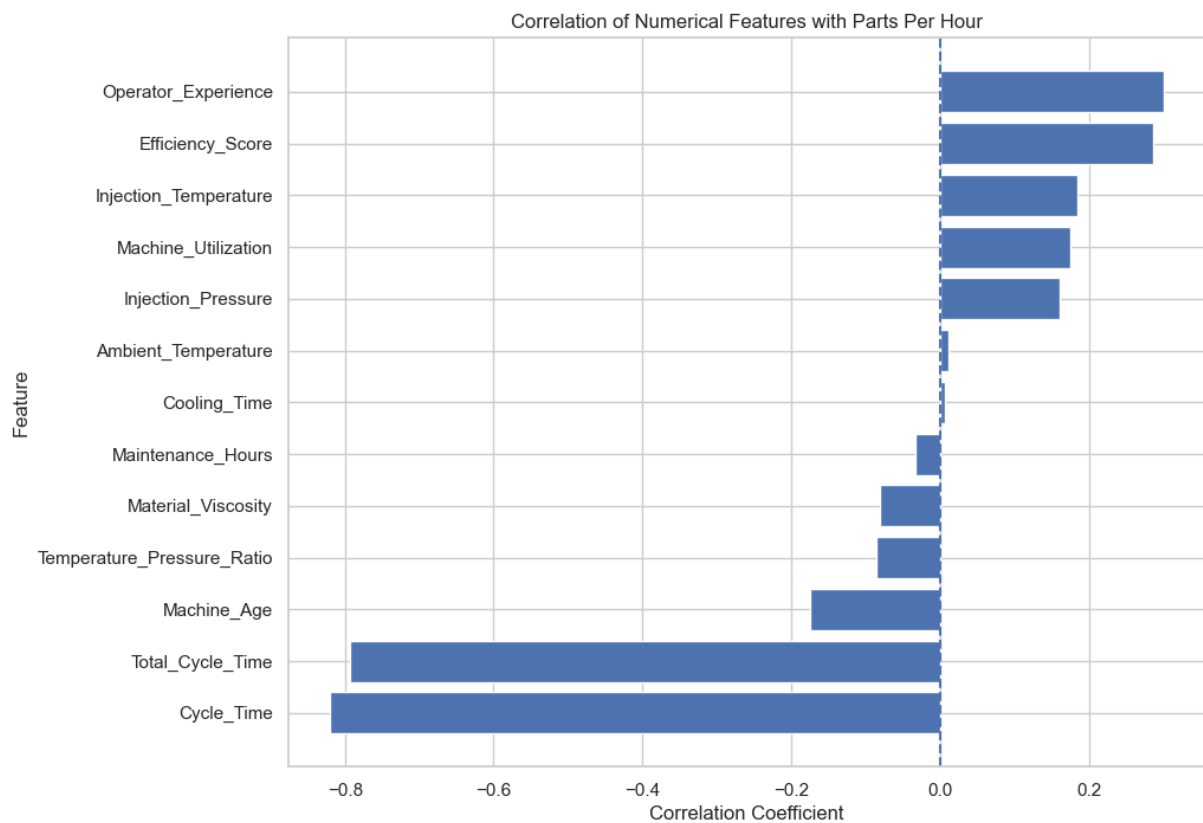


Figure 6. Correlation of each numerical feature with Parts_Per_Hour, sorted.

Key Numerical Observations

- **Cycle_Time** has the **strongest relationship** with output, a strong negative correlation of -0.819. Longer cycles mean fewer parts per hour.
- **Total_Cycle_Time** follows closely at -0.793. It combines cycle and cooling time, so it carries almost the same information as Cycle_Time.
- **Operator_Experience** and **Efficiency_Score** are the **leading positive factors**, both near +0.3.
- **Machine_Age** is **moderately negative** at -0.175, consistent with older machines producing less.
- **Ambient_Temperature**, **Cooling_Time** and **Maintenance_Hours** show **almost no linear relationship** with output on their own.

Multicollinearity note. Cycle_Time and Total_Cycle_Time are highly correlated with each other. Their individual regression coefficients can shift between the two and should be interpreted together rather than separately.

7. Operational Parameters vs Output

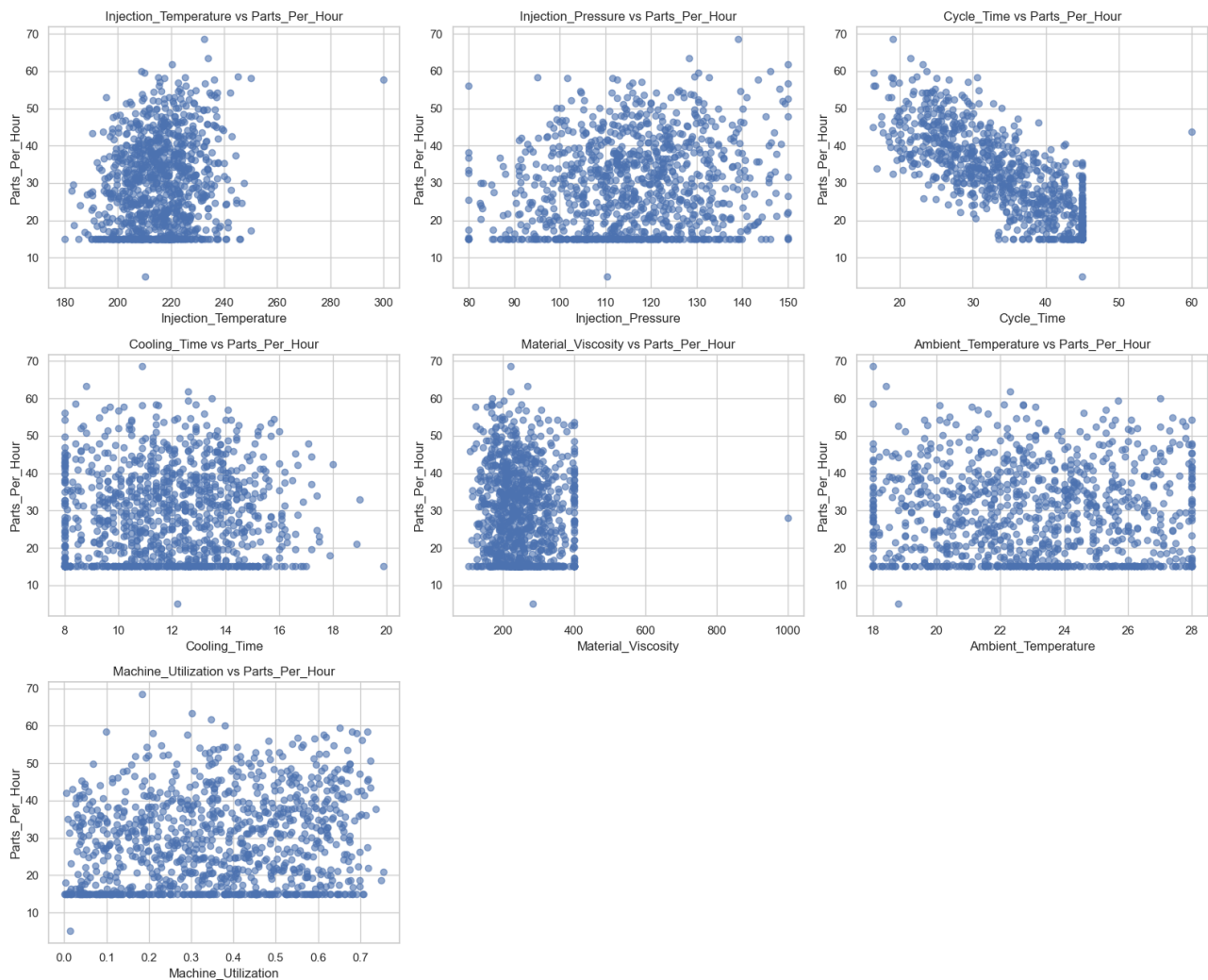


Figure 7. Scatter plots of seven key operating parameters against Parts_Per_Hour.

- **Cycle_Time shows a clear downward trend.** It is the only parameter with a strong visible linear pattern.
- **A horizontal band at 15 parts per hour appears in every panel.** Many records sit exactly at this level regardless of settings, which suggests a recorded minimum.
- **Isolated outliers are visible.** Examples include a temperature of 300, a viscosity of 1000 and a cycle time of 60.
- **The other parameters form wide clouds.** Their individual effects are weak and only become useful in combination.

8. Categorical Feature Analysis

Shift	n	Machine Type	n	Material Grade	n	Day of Week	n
Day	517	Type_A	384	Standard	591	Friday	168
Evening	277	Type_B	347	Premium	210	Wednesday	155
Night	206	Type_C	269	Economy	199	Thursday	150
						Saturday	141
						Tuesday	138
						Monday	136
						Sunday	112

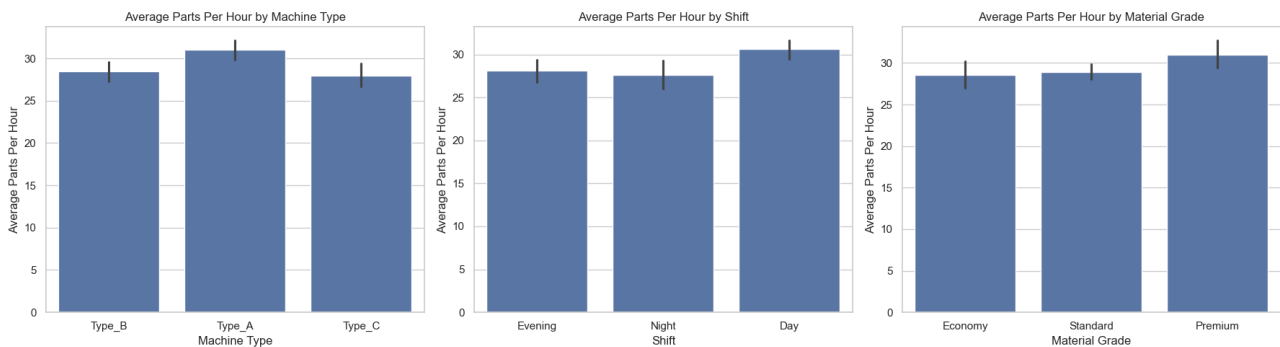


Figure 8. Average Parts_Per_Hour by machine type, shift and material grade, with 95% confidence intervals.

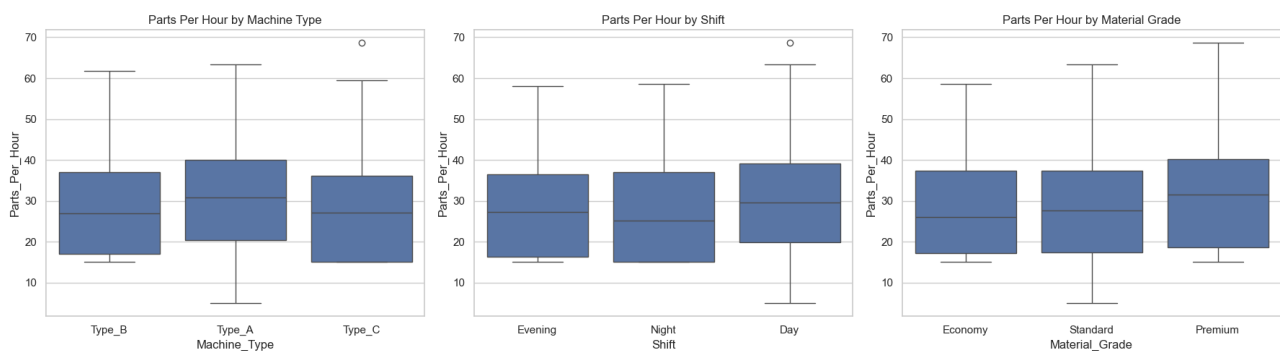


Figure 9. Distribution of Parts_Per_Hour by machine type, shift and material grade.

- **Type_A machines, the Day shift and Premium material show the highest averages**, each at roughly 31 parts per hour against about 28 for the other groups.
- **The boxes overlap heavily.** These differences are associations in observed data, not proof of cause.

9. Preprocessing and Modeling Methodology

9.1 Timestamp Processing

The Timestamp column was converted to datetime and the hour of day was extracted as a new numerical feature called **Hour**. The raw timestamp was then dropped because it is not a meaningful linear input. The dataset kept 19 columns after this step.

9.2 Features, Target and Leakage Control

The feature matrix X has 18 columns and the target y is Parts_Per_Hour. No feature was derived from the target, which avoids data leakage.

9.3 Train-Test Split

Set	Rows	Columns	Share
Training	800	18	80%
Testing	200	18	20%

The split used random_state = 42 so the results are reproducible.

9.4 Preprocessing Pipeline

Feature group	Columns	Steps
Numerical (14)	Injection_Temperature, Injection_Pressure, Cycle_Time, Cooling_Time, Material_Viscosity, Ambient_Temperature, Machine_Age, Operator_Experience, Maintenance_Hours, Temperature_Pressure_Ratio, Total_Cycle_Time, Efficiency_Score, Machine_Utilization, Hour	Median imputation, then StandardScaler
Categorical (4)	Shift, Machine_Type, Material_Grade, Day_of_Week	Most-frequent imputation, then OneHotEncoder with the first level dropped

A ColumnTransformer combines both branches into 26 model inputs. The dropped reference levels are the Day shift, Type_A, Economy grade and Friday. The whole preprocessor sits inside one Pipeline with the regressor, so it is fitted only on training data.

9.5 Model

An ordinary least squares **LinearRegression** model was fitted on the 800 training rows. The trained pipeline was saved to **manufacturing_output_model.pkl** with joblib for later deployment. The first predictions produced by the fitted model are shown below.

Sample	1	2	3	4	5
First training predictions	43.77	41.58	15.52	23.04	50.17
First testing predictions	19.94	44.43	34.09	39.68	18.35

10. Model Performance Evaluation

Dataset	R ²	MSE	RMSE	MAE
Training	0.924	11.056	3.325	2.590
Testing	0.906	12.315	3.509	2.716
Gap (Test - Train)	-0.018	+1.259	+0.184	+0.126

- **R² of 0.906** means the model explains about 91% of the variation in test-set output.
- **MAE of 2.716** means predictions miss by about 2.7 parts per hour on average, roughly 9% of mean output.
- **The small train-test gap** shows that the model generalizes well and is not overfitting.

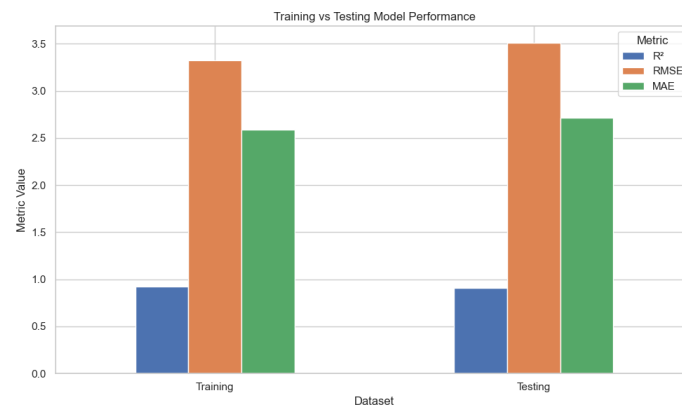


Figure 10. Training vs testing R², RMSE and MAE.

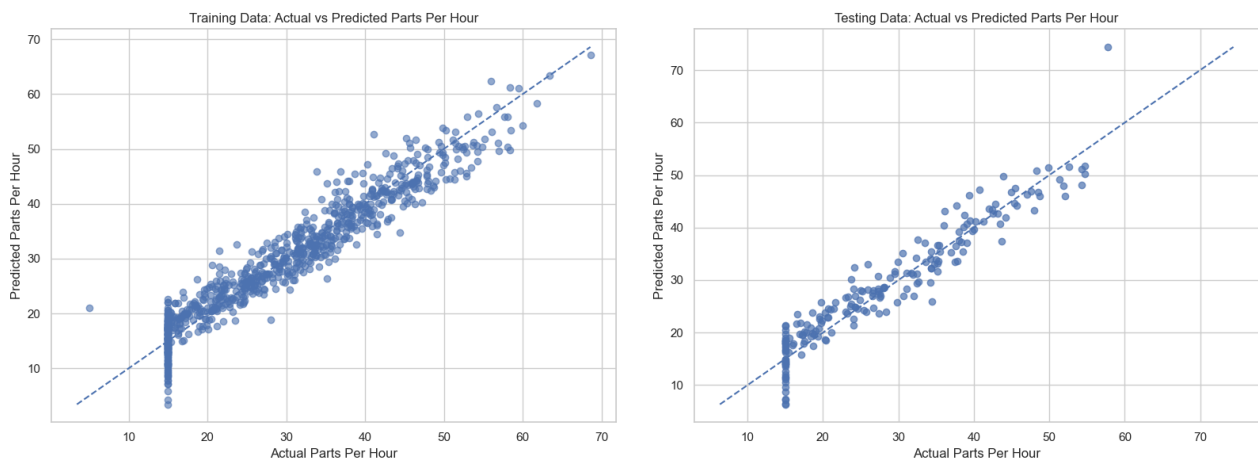


Figure 11. Actual vs predicted Parts_Per_Hour for training (left) and testing (right) data. Points close to the dashed line are accurate predictions.

11. Residual Analysis

A residual is the actual value minus the predicted value. The first ten rows of each set are shown below.

Training sample

#	Actual	Pred.	Resid.
0	39.8	43.766	-3.966
1	41.5	41.578	-0.078
2	15.0	15.517	-0.517
3	19.7	23.037	-3.337
4	45.7	50.174	-4.474
5	24.7	24.280	0.420
6	26.5	23.798	2.702
7	52.7	50.523	2.177
8	47.8	46.124	1.676
9	38.9	35.220	3.680

Testing sample

#	Actual	Pred.	Resid.
0	18.6	19.936	-1.336
1	42.8	44.429	-1.629
2	34.2	34.088	0.112
3	40.0	39.681	0.319
4	15.0	18.349	-3.349
5	15.0	20.414	-5.414
6	33.6	33.501	0.099
7	15.0	14.092	0.908
8	25.0	28.021	-3.021
9	24.2	32.371	-8.171

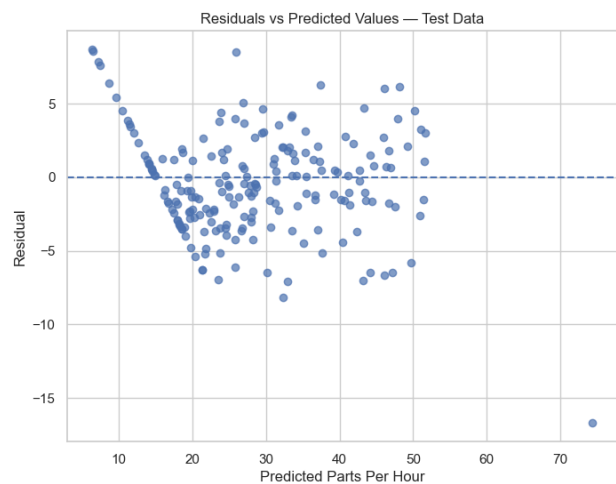
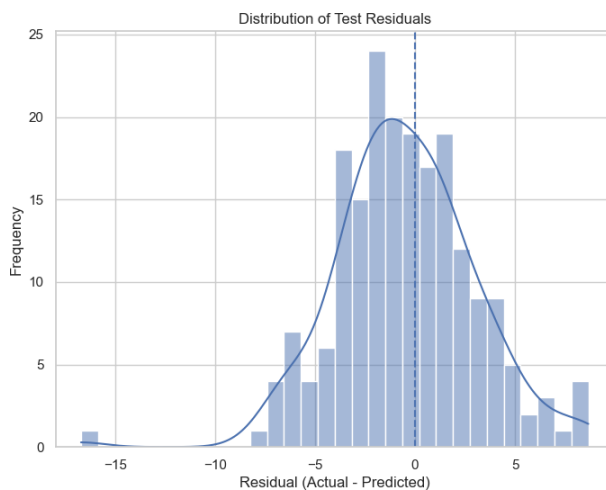


Figure 12. Distribution of test residuals (left) and residuals vs predicted values (right).

- **Most errors are centred near zero** and fall between about -7 and +8 parts per hour.
- **A systematic pattern appears at low predictions.** Below about 15 predicted parts per hour, residuals climb steadily. The model predicts values down to about 6, but actual output is recorded at 15, so a straight-line model cannot follow the floor.
- **One large outlier** was predicted at about 74 against an actual of about 58, a residual near -16.7.

12. Feature Coefficients

Feature	Coef.	Feature	Coef.
Total_Cycle_Time	-10.059	Operator_Experience	+1.027
Shift_Night	-3.961	Day_of_Week_Thursday	+0.944
Machine_Type_Type_C	-3.920	Material_Viscosity	-0.942
Material_Grade_Premium	+3.106	Temperature_Pressure_Ratio	-0.934
Efficiency_Score	+2.866	Day_of_Week_Tuesday	+0.754
Cooling_Time	+2.669	Day_of_Week_Monday	+0.631
Machine_Type_Type_B	-2.422	Day_of_Week_Saturday	+0.566
Injection_Temperature	+2.303	Cycle_Time	-0.565
Machine_Age	-2.201	Machine_Utilization	-0.400
Day_of_Week_Wednesday	+1.752	Day_of_Week_Sunday	+0.147
Shift_Evening	-1.742	Ambient_Temperature	-0.121
Material_Grade_Standard	+1.644	Maintenance_Hours	-0.072
Injection_Pressure	+1.168	Hour	+0.035

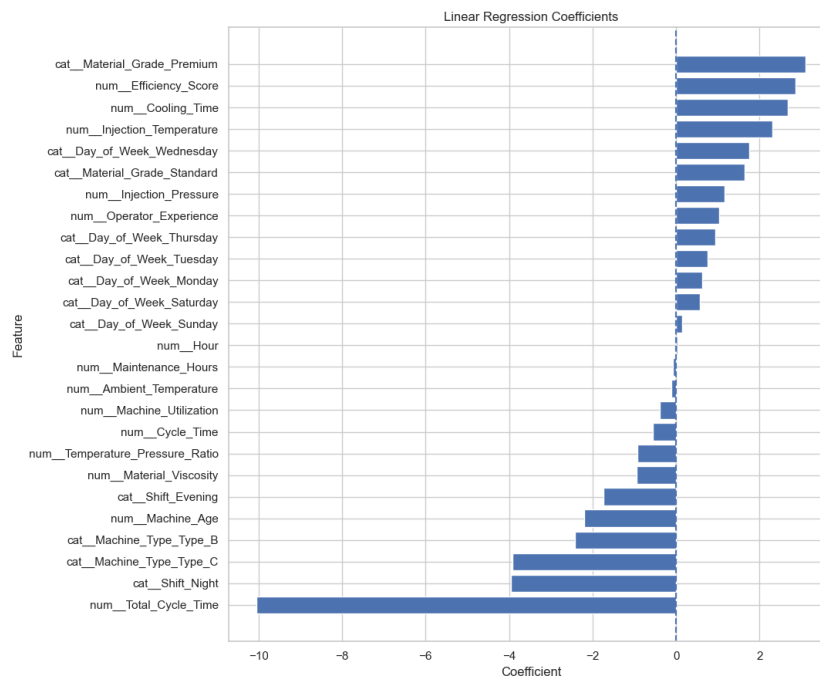


Figure 13. Linear Regression coefficients for all 26 model inputs.

Numerical coefficients show the change in predicted output for a one-standard-deviation increase in that feature, holding the others constant. Categorical coefficients are differences from the dropped reference level. **Total_Cycle_Time dominates at -10.06 parts per hour per standard deviation.** Night shift and Type_C machines are each associated with about 4 fewer parts per hour than the Day shift and Type_A. Premium material is associated with about 3.1 more parts than Economy. Cycle_Time itself gets only -0.565 because Total_Cycle_Time absorbs most of their shared effect. These are model associations, not causal effects.

13. Insights, Recommendations and Conclusion

13.1 Key Manufacturing Insights

1. **Cycle duration is the main driver of output.** Cycle_Time and Total_Cycle_Time have the strongest negative relationships with Parts_Per_Hour.
2. **Operator experience and efficiency are positively associated with output.**
3. **Output differs by machine type, shift and material grade,** although the distributions overlap.
4. **The model predicts unseen data closely,** with a test R^2 of 0.906 and an MAE of 2.716.
5. **Errors are mostly balanced around zero,** except for the low-output floor near 15 parts per hour.

13.2 Production Optimization Recommendations

1. **Monitor cycle duration.** Investigate unusually long cycle times, as they are tied to lower output.
2. **Track machine-level performance.** Compare machines on consistent KPIs and review Type_C units.
3. **Review shift-level output.** Night and Evening shifts trail the Day shift and may need operational review.
4. **Review material-grade performance.** Check process stability across grades before changing settings.
5. **Use the model for production planning.** Predicted output can support scheduling and throughput estimates.
6. **Flag large prediction errors.** Big gaps between actual and predicted output are signals for investigation.

13.3 Limitations and Next Steps

- **Handle the output floor.** Clip predictions at 15 or test a model that respects the floor, such as a tree-based or censored regression model.
- **Treat extreme values.** Review the anomalous maxima in maintenance hours, viscosity and temperature.
- **Reduce redundancy.** Consider keeping only one of Cycle_Time and Total_Cycle_Time for clearer coefficients.
- **Deploy the saved model.** Serve manufacturing_output_model.pkl through FastAPI and Docker as the next stage.

Conclusion. The Linear Regression pipeline predicts hourly injection molding output with high accuracy and generalizes well from training to testing data. Cycle time is the most important lever for throughput, and the saved model is ready for the deployment stage.